RE07 Global Infrastructure Investment and Management

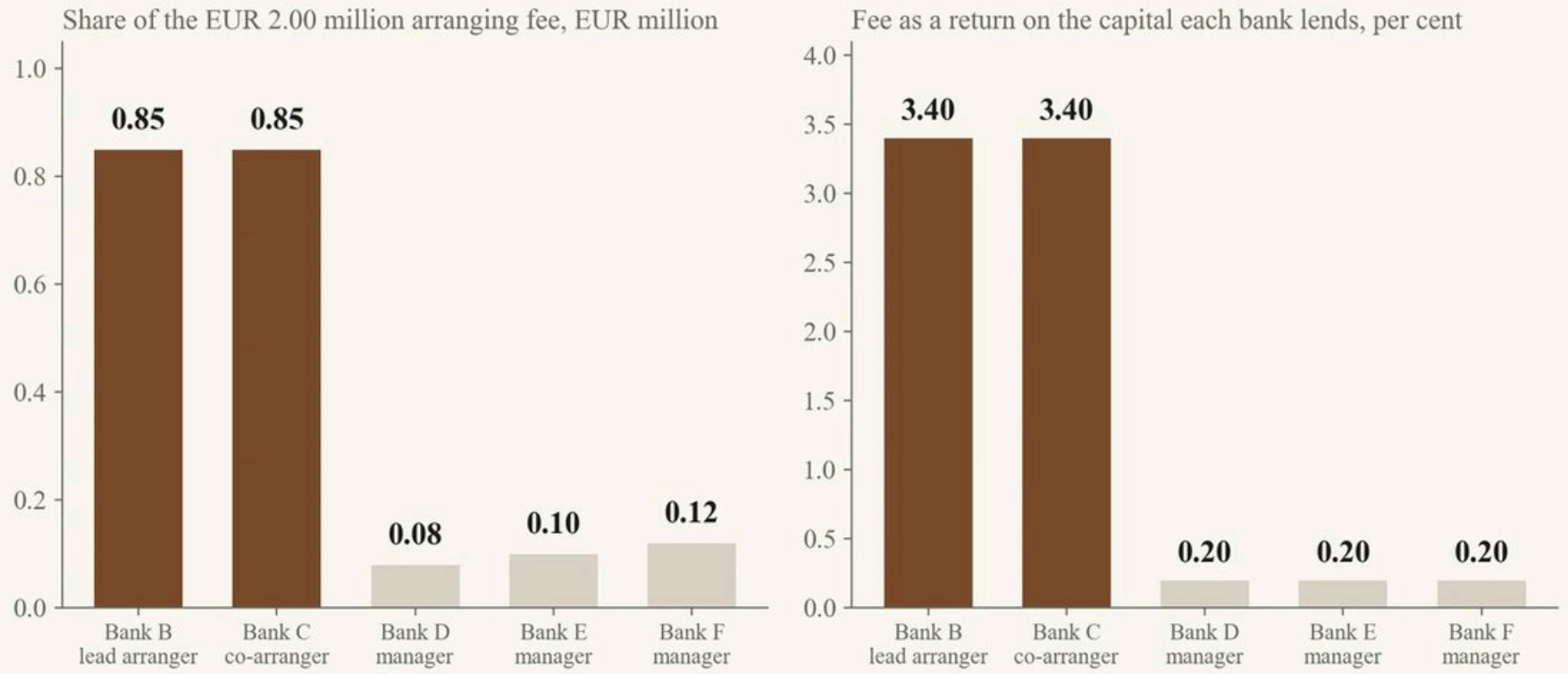
5. Intermediation and Deal Brokerage

Helen Bao
Department of Land Economy, University of Cambridge

Lecture Outline

Arranging and certifying a financing
Brokerage, advice and the selection of managers
Private credit and the data-centre boom
Three models of what intermediaries sell
Case Study: Topic to Be Agreed (Justin Chiu, CK Asset Holdings)

The Fee Stack on a Syndicated Loan



Source: Gatti (2024), Project Finance in Theory and Practice, Tables 6.8 to 6.10, printed pages 207 and 208. Illustrative loan of EUR 200 million; Banks B and C lend EUR 25 million each, Banks D, E and F lend EUR 40, 50 and 60 million.

What the sponsor pays: EUR 1.5 million to the adviser and EUR 2.0 million to the arranger on a loan of EUR 200 million, 1.75 per cent before the first drawdown

Who keeps it: the two lead banks keep EUR 0.85 million each and the three managers EUR 0.08 to 0.12 million

The return on capital: 3.4 per cent on the EUR 25 million each lead bank finally lends, 0.2 per cent for managers who lend EUR 40 to 60 million

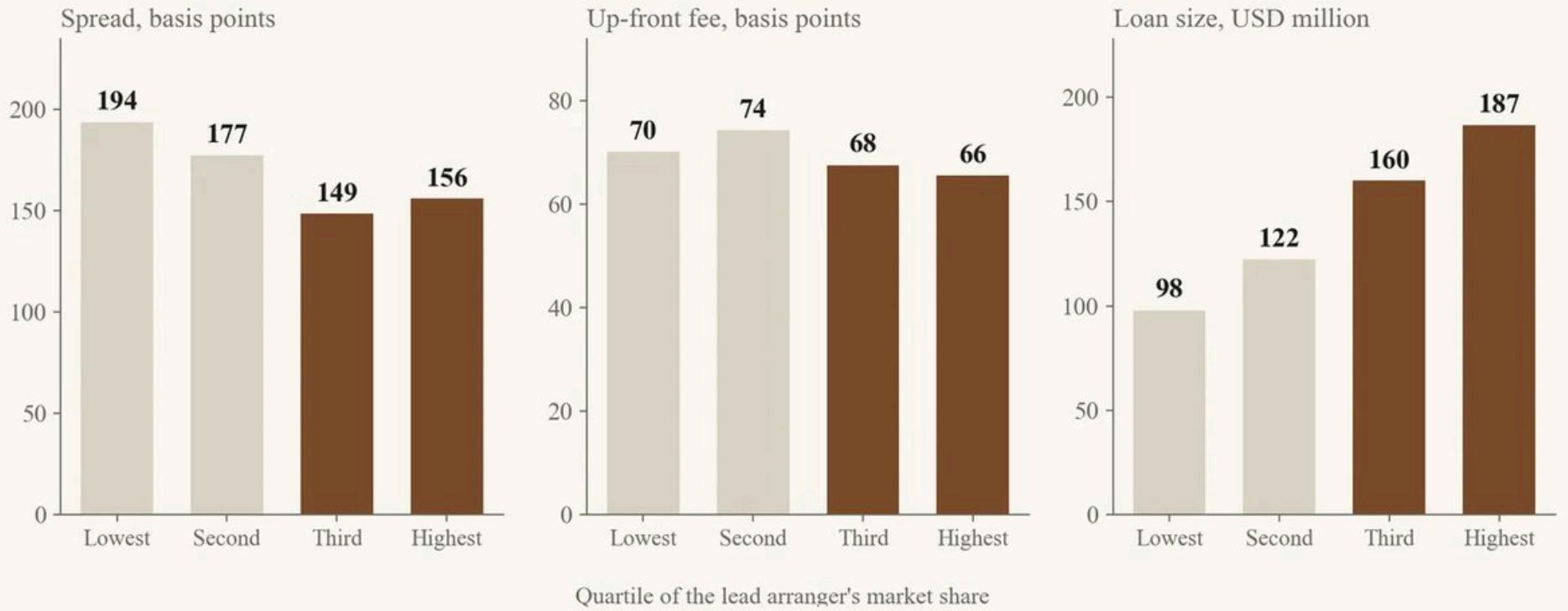
The incentive: the adviser's fee is a share of the debt, so it rises with the leverage the adviser recommends

Underwriting, Club Deals and the Debt Fund

Structure	How it works	What the text records (printed page)
Two-stage syndication	A small group underwrites, then resells part of the loan; the arranger keeps a final take of around 10 per cent	One to two months, up to twelve for very large deals (192)
Club deal	Four to six banks provide the whole loan; fees are shared equally	Quicker, and shields the banks from changes in the market (193)
Partnership with an institution	An insurer co-invests in loans a bank originates; the bank keeps a share	Crédit Agricole kept 20 per cent of EUR 1 billion transferred in 2012 (195)
Securitisation	Loans pooled and sold as rated notes	Project loan CLOs and CDOs almost disappeared after 2008 (195 to 196)
Debt fund	An institution commits to a fund that originates the loans	170 funds reached close from 2010 to 2020, raising about USD 88 billion (196)

Gatti (2024), Project Finance in Theory and Practice, sections 6.2.1 and 6.2.2, printed pages 191 to 196. Debt fund figures are for 2010 to 2020.

Arranger Prestige and Loan Terms



Source: Gatti, Kleimeier, Megginson and Steffanoni (draft of 19 March 2007), Table 3 Panel A, page 21; abstract. Sufi (2007), Journal of Finance 62(2), abstract.

The sample: 4,122 project finance loans worth USD 584 billion, arranged from 1991 to 2005

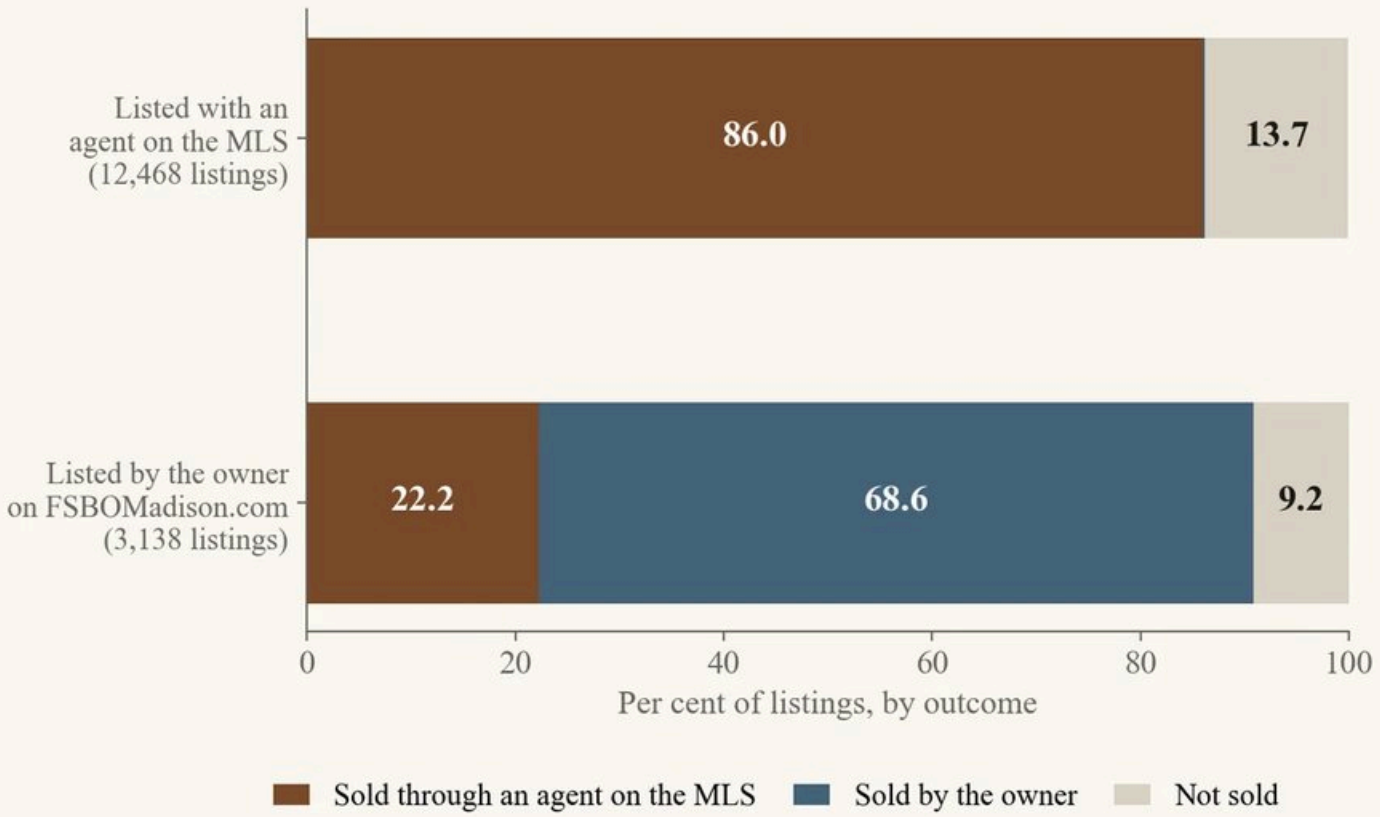
Spread: 194 basis points where the lead arranger's market share is lowest, 156 where it is highest

Fee: no higher for the larger arrangers: 70 and 66 basis points, so prestige is not paid for directly

Size: larger loans, USD 98 million against 187 million: the arranger's name buys capacity

Retained share: Sufi (2007): the lead bank keeps more of a loan that needs more monitoring

Selling a House without an Agent



The comparison: Madison, Wisconsin, 1998 to 2004: 3,138 homes listed by the owner on a website, 12,468 listed with an agent

Success: 86.0 per cent of agent listings sold through the agent, 68.6 per cent of owner listings sold by the owner; 22.2 per cent of owner listings ended with an agent

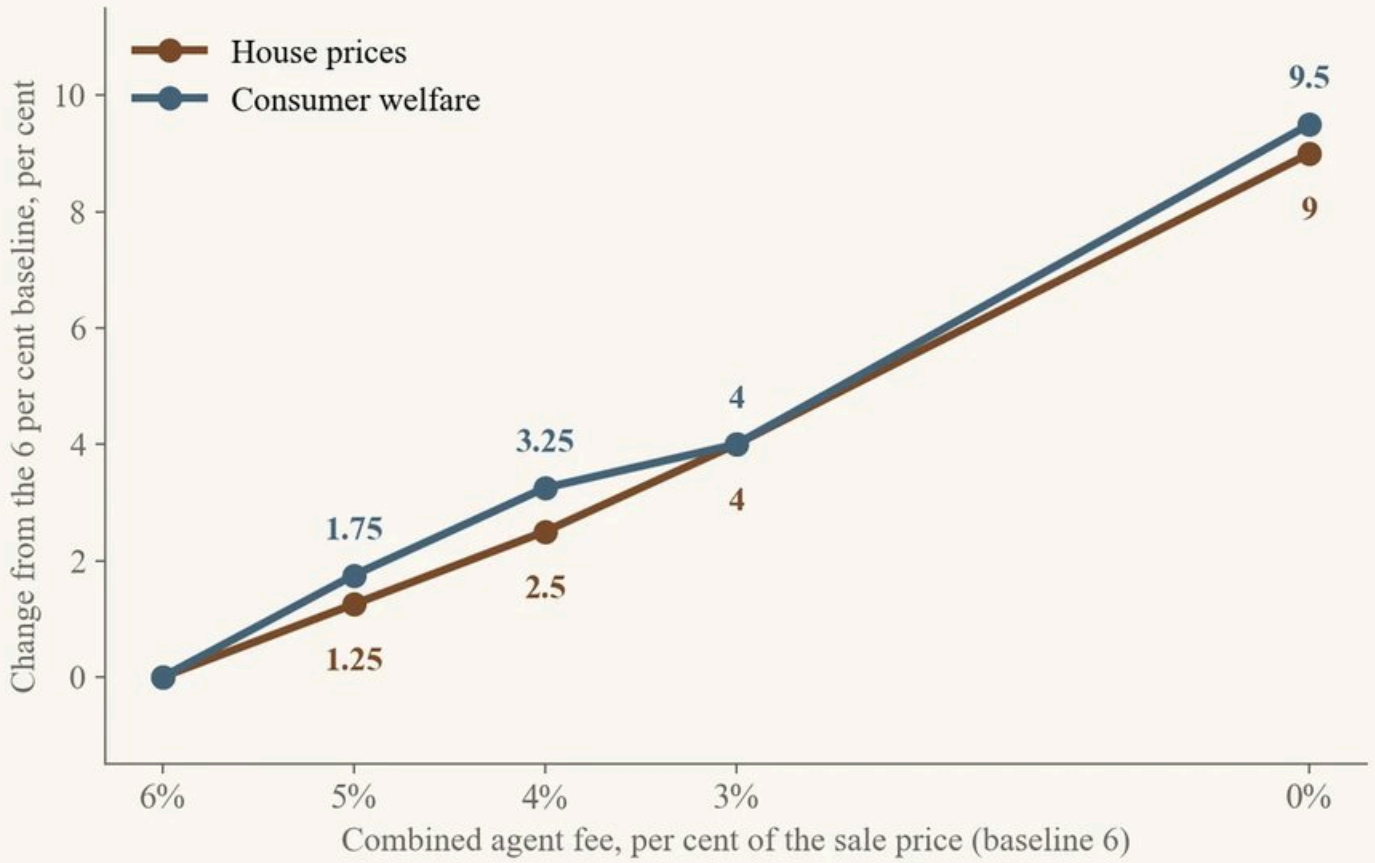
Price: no support for a higher price through the agent; the premium for selling by the owner was about 3 per cent after controls

Time: listing with an agent shortens the time to sell

The reading: the agent sells speed and a better chance of a sale, not a better price

Hendel, Nevo and Ortalo-Magné (2007), NBER Working Paper 13360, Table 1 (page 29) and pages 14 to 15; abstract. Published in the American Economic Review 99(5), 2009.

Agents' Fees after the 2024 Settlement



The institution: the seller pays both agents, typically 5 to 6 per cent of the price; the buyer agent's 3 per cent was posted on the MLS until August 2024

The prediction: a calibrated model says that cutting the combined fee from 6 to 4 per cent raises house prices by 2.5 per cent, because resale becomes cheaper

Who gains: consumer welfare rises by 3.25 per cent, but most of it accrues to current owners

Another account: Grochulski and Wang: a cost-based fee system would add about USD 38.5 billion a year, mostly by cutting agents' rents

Buchak, Matvos, Piskorski and Seru (2025), NBER Working Paper 32855, pages 16 to 17 and Figure 1; Grochulski and Wang (2025), Richmond Fed Working Paper 24-01R. Model results, not outcomes after the settlement.

Investment Consultants as Gatekeepers

Their role: advise institutions on which managers to hire; about USD 25 trillion of assets was advised on in June 2011

Influence: winning the recommendation of one third of consultants brings about 10 per cent, USD 0.8 billion, within a year

Performance: recommended funds earned about 1 per cent less than others, equally weighted; none outperformed once their size is allowed for

What consultants may add: reassurance, and cover for a sponsor that has to defend its choice

The limit: US active equity only; the authors say other asset classes may differ

Jenkinson, Jones and Martinez (September 2014 draft), pages 1 to 6. Published in the Journal of Finance, which I recall as volume 71 (2016).

Private Credit and the AI Boom

Measure	Figure given in the Bulletin	Where (printed page)
Assets of private credit funds	About USD 100 billion in 2010, over USD 2.2 trillion now	3, footnote 4
Direct loans to AI-related firms outstanding	From near zero to over USD 200 billion	4, Graph 3.A
Share of private credit loans going to them	From under 1 per cent to almost 8 per cent	4, Graph 3.A
Originations to them in 2025	Over USD 40 billion, against about USD 3 billion in 2010	4, footnote 5
Loan spreads against non-AI loans	Close; the difference is insignificant after controls	5, Graph 3.C, footnote 6

Funds lend directly and hold the loan to maturity. The Bulletin warns that some structures move leverage off the balance sheet, and notes circular financing within the AI ecosystem (page 4).

Aldasoro, Doerr and Rees (2026), BIS Bulletin No 120, 7 January 2026, pages 3 to 5. 'AI firms' are PitchBook's Artificial Intelligence, Big Data and Cloud Tech verticals.

Three Models of What Intermediaries Sell

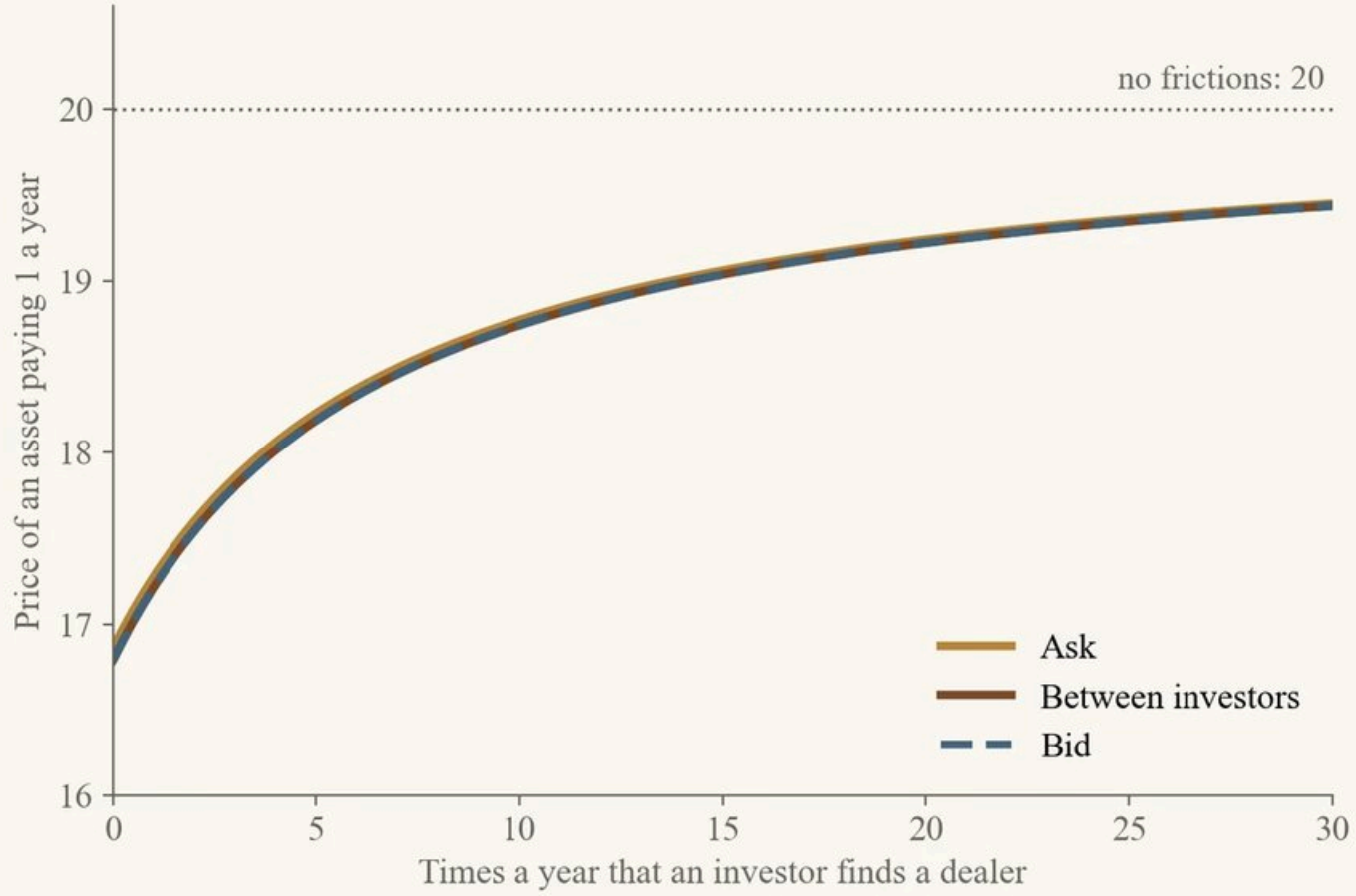
Immediacy: Duffie, Gârleanu and Pedersen (2005): a dealer is paid for shortening a search

Skill: Berk and Green (2004): a manager is paid for a talent that investors cannot see and that competition leaves with the manager

Assurance: Franzoni (1999): a certifier is paid for an approval that others rely on, and liability keeps it honest

The test: for any intermediary, say which of the three it sells, what it is paid, and who bears the loss if it fails

Duffie et al. (2005): Search and the Dealer



Investors: high or low type, owning or not; a low-type owner wants to sell and a high-type non-owner wants to buy

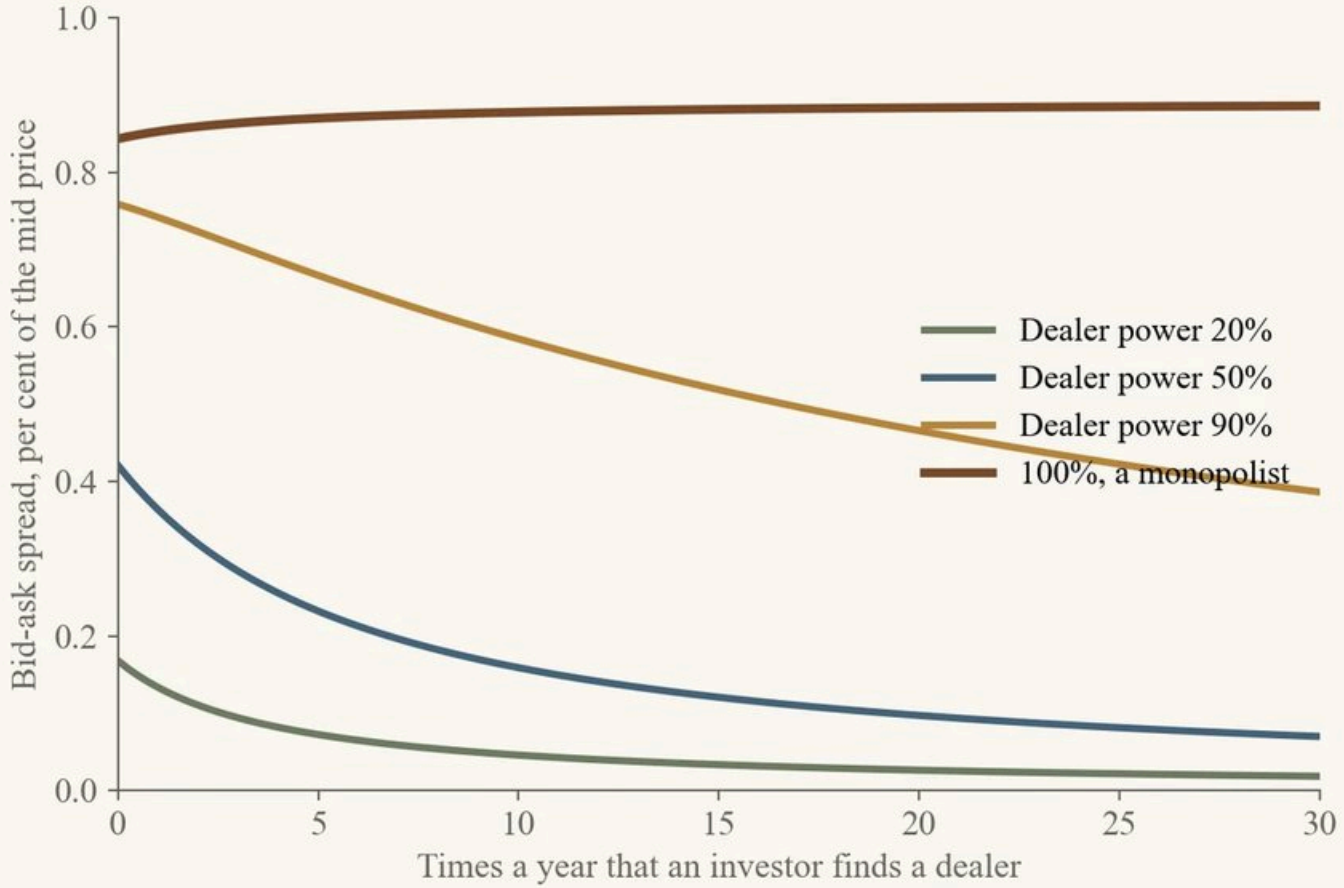
Search: they meet at random, and a dealer can be found too, who takes either side and holds no inventory

Prices: each is the value of the income less a discount for waiting

At the generic settings: with no dealer the discount is 16.0 per cent and the wait 16 months; with a dealer found twice a year, 12.3 per cent and 4.5 months

Duffie, Gârleanu and Pedersen (2005), NBER Working Paper 10816, pages 5 to 13; Theorem 2, equations 11 to 13. Generic settings of the interactive page.

The Cost of Immediacy



The spread falls with reach: from 0.32 per cent at two dealer contacts a year to 0.10 at twenty, for a dealer with half the power

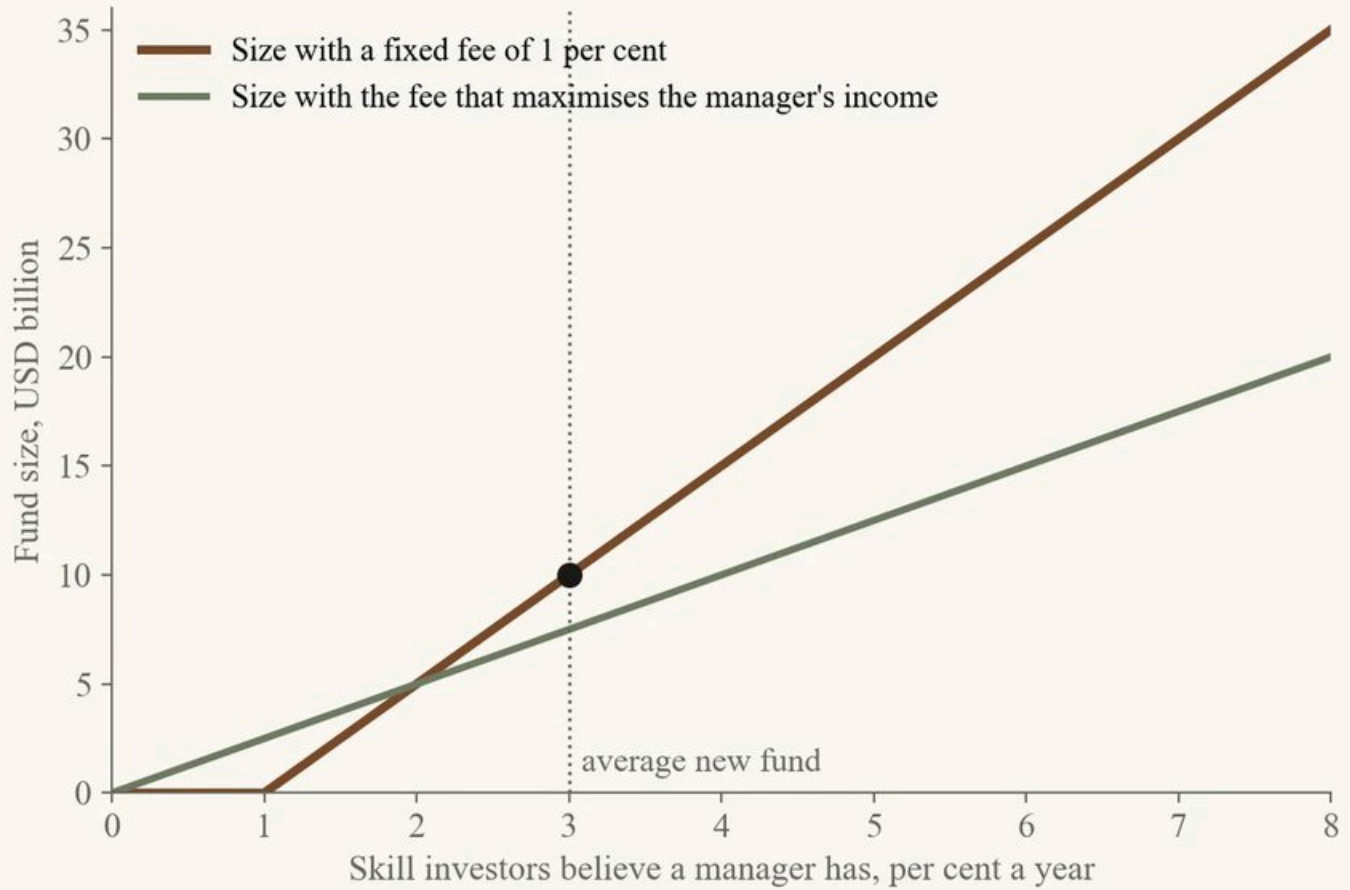
A monopolist: the spread rises with reach, from 0.86 to 0.88 per cent

The discount is larger: a 12.3 per cent discount against a 0.32 per cent spread; the dealer shortens the wait but does not remove it

An unlisted stake: dealers found every two years and few buyers: a wait of 17 months and a discount of 16 per cent

Duffie, Gârleanu and Pedersen (2005), NBER Working Paper 10816, Theorem 2 and pages 13 to 14. Generic settings of the interactive page (dealer-spread.html); not calibrated to any asset.

Berk and Green (2004): Skill, Size and Rents



Managers: differ in skill, which no one observes; everyone learns it from returns

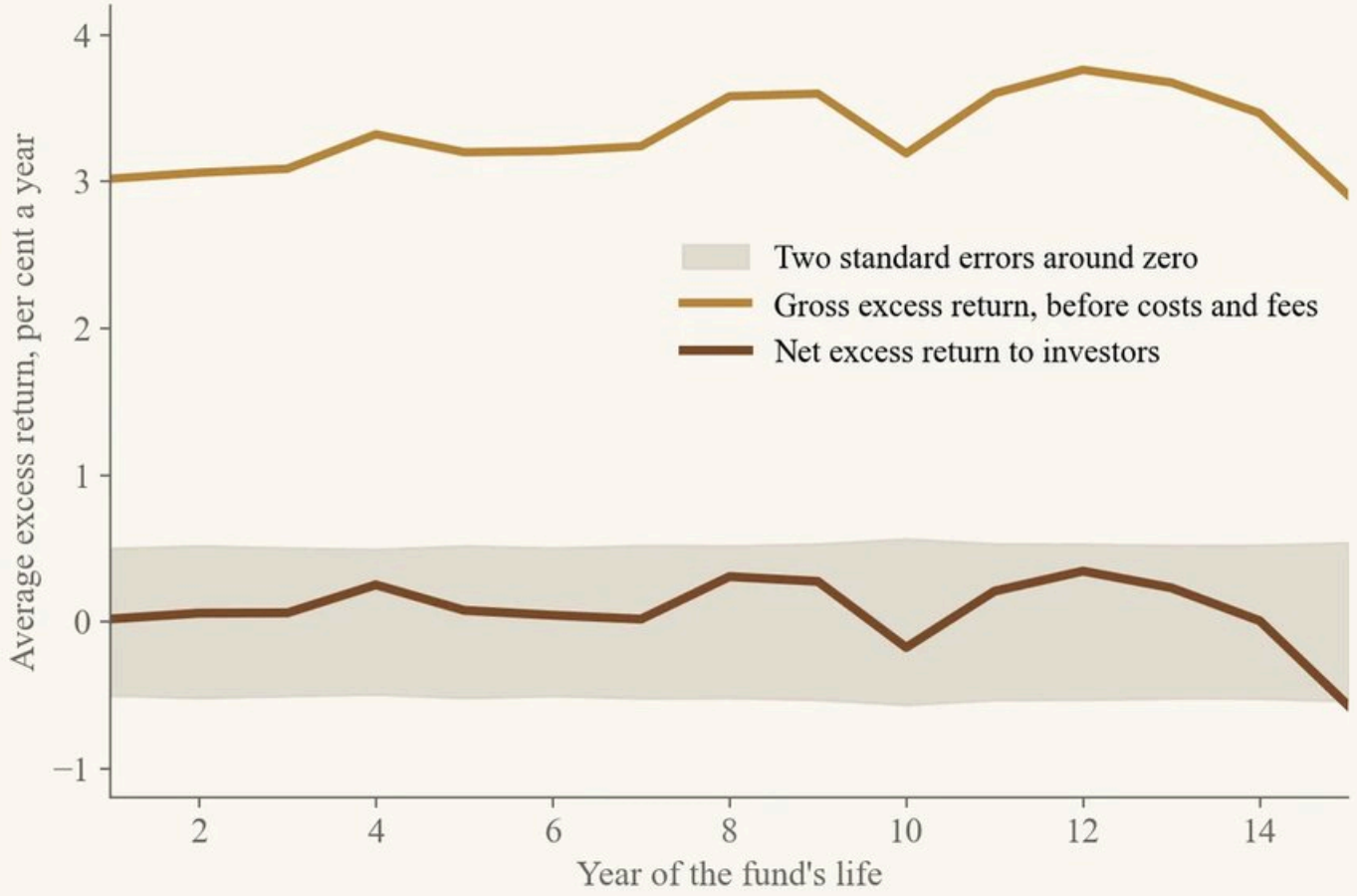
Investors: put money in until the expected excess return after costs and fees is zero

Size: a fund believed to add 3 per cent a year, with a fee of 1 per cent and a cost of size of 0.4 points a billion, grows to 10 billion dollars

The rent: the manager collects 100 million dollars a year in fees and managing the fund costs 200 million; investors earn nothing above the benchmark

Berk and Green (2004), Journal of Political Economy 112(6), equations 3, 4, 9 and 10, pages 1275 to 1277. Quadratic cost and generic settings are mine (fund-rents.html).

Returns and Flows in a Population of Funds



Gross: the survivors earn about 3 per cent a year before costs and fees

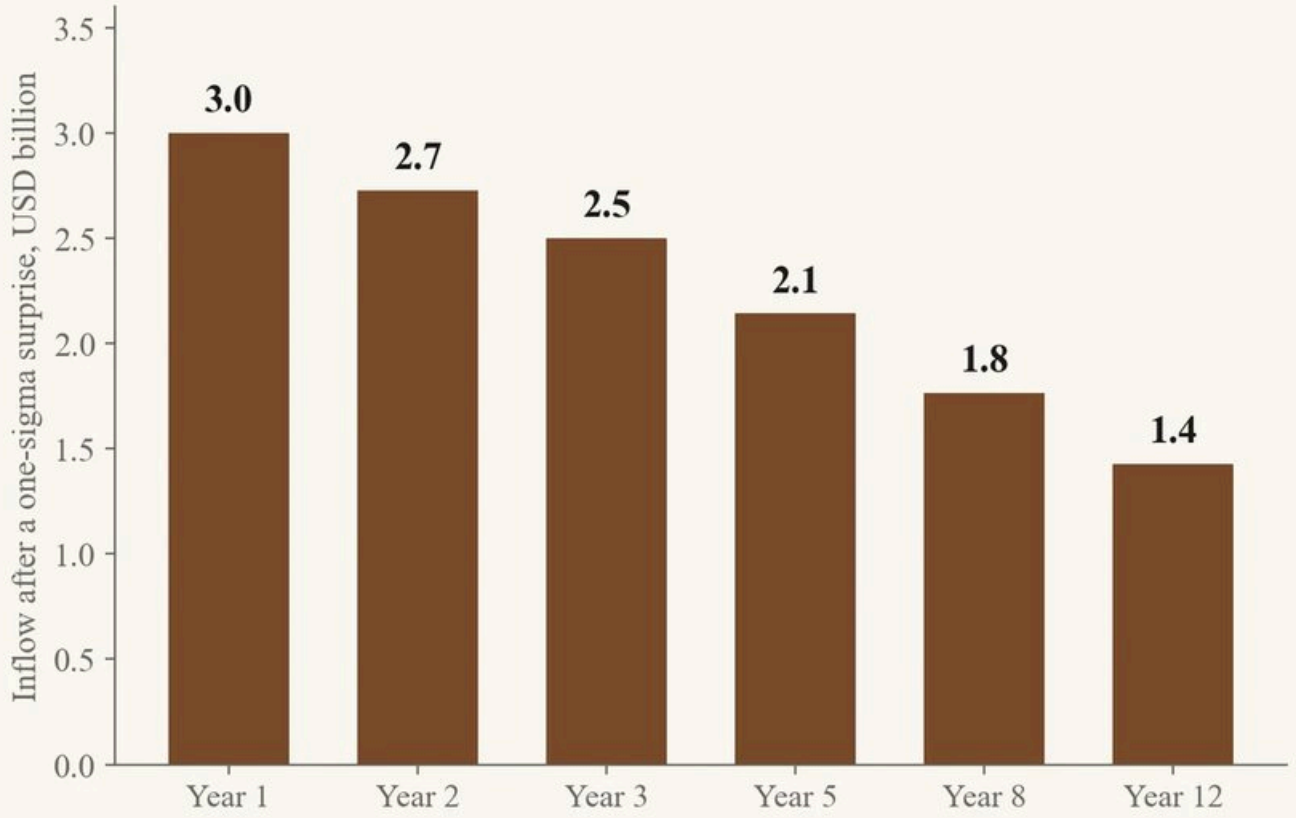
Net: investors earn 0.08 per cent on average, inside two standard errors of zero, and last year's net return does not predict this year's (correlation 0.01)

The inference: the response of money to performance and the absence of persistence are what rational investors would produce

The evidence on skill: Berk and van Binsbergen: US mutual fund managers add dollars, about USD 2 million a year on average, and the skill persists for up to ten years

Berk and Green (2004), pages 1275 to 1276; Berk and van Binsbergen (2012), NBER Working Paper 18184, abstract. Simulation of 600 funds at the generic settings of fund-rents.html.

Flows after a Good Year



The weight on the latest return: falls with the length of the record, from 0.10 in year 1 to 0.05 in year 12

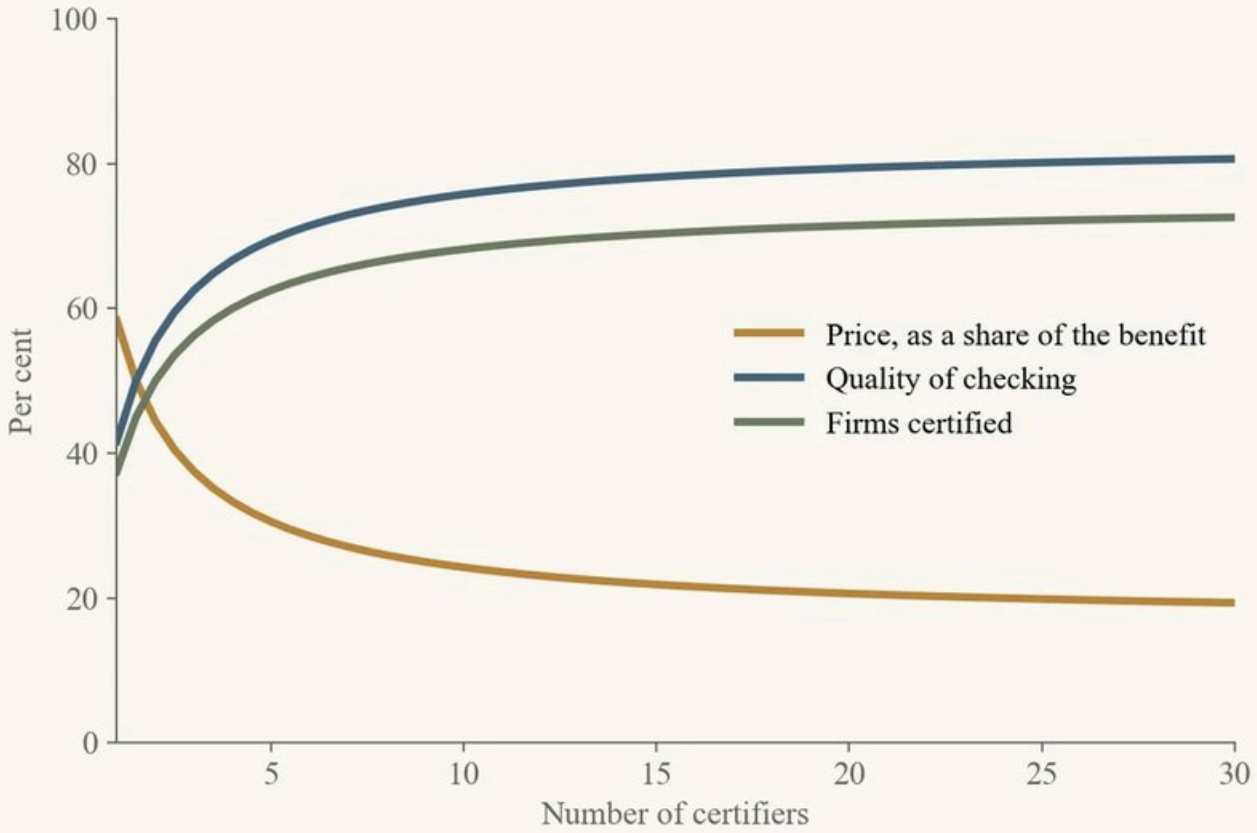
The inflow: after a one-standard-deviation surprise, 3.0 billion dollars in year 1 and 1.4 in year 12

Consultants: a recommendation from one third of them moves a fund by about 10 per cent in a year (slide 8)

Private funds: lock capital in for years, so the response comes at the next fund, which the model does not describe

Berk and Green (2004), equations 5 and 6, page 1275, with a quadratic cost; generic settings of fund-rents.html.

Franzoni (1999): Certification and Liability



Firms: need a certificate to enter; compliance costs are spread evenly, and only firms with costs below the benefit should enter

Certifiers: compete on the number of certificates, and check hard only if the chance of discovery times the penalty exceeds the cost of checking

Price: 0.20 with four certifiers against a cost of checking of 0.10, 0.12 with thirty

Quality: the share of non-compliant applicants caught rises from 42 per cent with one certifier to 81 per cent with thirty

Franzoni (1999), University of Bologna working paper, pages 6 to 14. Generic settings of certification.html; not calibrated to any market.

Liability and the Value of Certification



Weak liability: at a penalty of 0.3 every firm applies and the system is worth minus 16 per cent of the best outcome, less than nothing

The generic penalty: 0.6 gives 52 per cent

A large penalty: a penalty of 6 gives 73 per cent and no more however large it becomes, because the fee keeps out some compliant firms

Competition: 41 per cent with one certifier, 52 with four and 53 with thirty

Franzoni (1999), welfare on pages 17 to 18 and equation 8. Generic settings of certification.html; not calibrated to any market.

Three Models Compared

	Duffie, Gârleanu and Pedersen (2005)	Berk and Green (2004)	Franzoni (1999)
What is sold	Immediacy: a shorter search	A skill investors cannot see	An approval others rely on
What disciplines it	The next dealer or the next investor	Competition among investors for the manager's output	Liability if the approval proves wrong
What the price reflects	Search frictions and bargaining power	Skill and the cost of size	The number of certifiers and the liability
Who gets the rent	The dealer, if it has most of the power	The manager, not the investors	The certifiers, when they are few
A case could test it by	The time to sell and the discount to value	Fund size, fees and net returns	Ratings, liability terms and defaults

Duffie, Gârleanu and Pedersen (2005); Berk and Green (2004); Franzoni (1999). Interactive pages in web/.

Case Study: Topic to Be Agreed (Justin Chiu, CK Asset Holdings)

The business and its position: what the group owns, and how it is organised

The transaction: what was bought, sold or financed, and why

The intermediaries: who advised, arranged, rated and lent, and how each was paid

The holding and its value: how the assets are held, reported and valued

Discussion: questions for the guest

Summary

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